



Interest Rate Spillovers to the UK Gilt Curve

VAR Analysis of Cross-Market Shock Transmission
US & EU → UK Level, Slope, Curvature

Rates Research | 5 May 2026

Methodology: VAR on Nelson-Siegel Factors

We decompose the UK, US, and Euro-area government yield curves into three latent factors via the Nelson-Siegel model:

$$y(\tau) = \underbrace{\beta_1}_{\text{Level}} + \underbrace{\beta_2 \cdot \frac{1 - e^{-\lambda\tau}}{\lambda\tau}}_{\text{Slope}} + \underbrace{\beta_3 \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right)}_{\text{Curvature}}$$

where $\lambda = 0.10$ (optimised by RMSE). Factors are extracted daily from 6 maturities (1Y–20Y) for each region, then resampled to weekly frequency. A 9-variable VAR(1) is estimated on:

Cholesky ordering: US_{Level} → US_{Slope} → US_{Curv} → EU_{Level} → EU_{Slope} → EU_{Curv} → UK_{Level} → UK_{Slope} → UK_{Curv}

Rationale: US rates are most exogenous (global reserve currency); UK is the “receiver” we focus on.

Key VAR Coefficients (UK equations)

Regressor (lag-1)	UK Level	UK Slope	UK Curvature
US Level	—	—	—
US Slope	+0.074** (via US eq)	—	—
EU Level	+0.075** (via US eq)	—	—
Own lag	0.989***	0.946***	0.939***

*High persistence (>0.93) in all UK factors confirms near-unit-root dynamics.
Cross-border transmission operates mainly through contemporaneous correlation.*

Granger Causality (at 5% level)

Causing block	→ UK Level	→ UK Slope	→ UK Curvature
US (Level+Slope+Curv)	Significant ($p=0.048$)	Not sig. ($p=0.35$)	Significant ($p=0.022$)
EU (Level+Slope+Curv)	Not sig. ($p=0.89$)	Not sig. ($p=0.91$)	Not sig. ($p=0.59$)

Interpretation: US rates *predictively lead* UK Level and Curvature with a 1-week lag. EU rates do *not* Granger-cause UK factors—their influence is purely contemporaneous (same-day co-movement captured in the residual correlation structure).

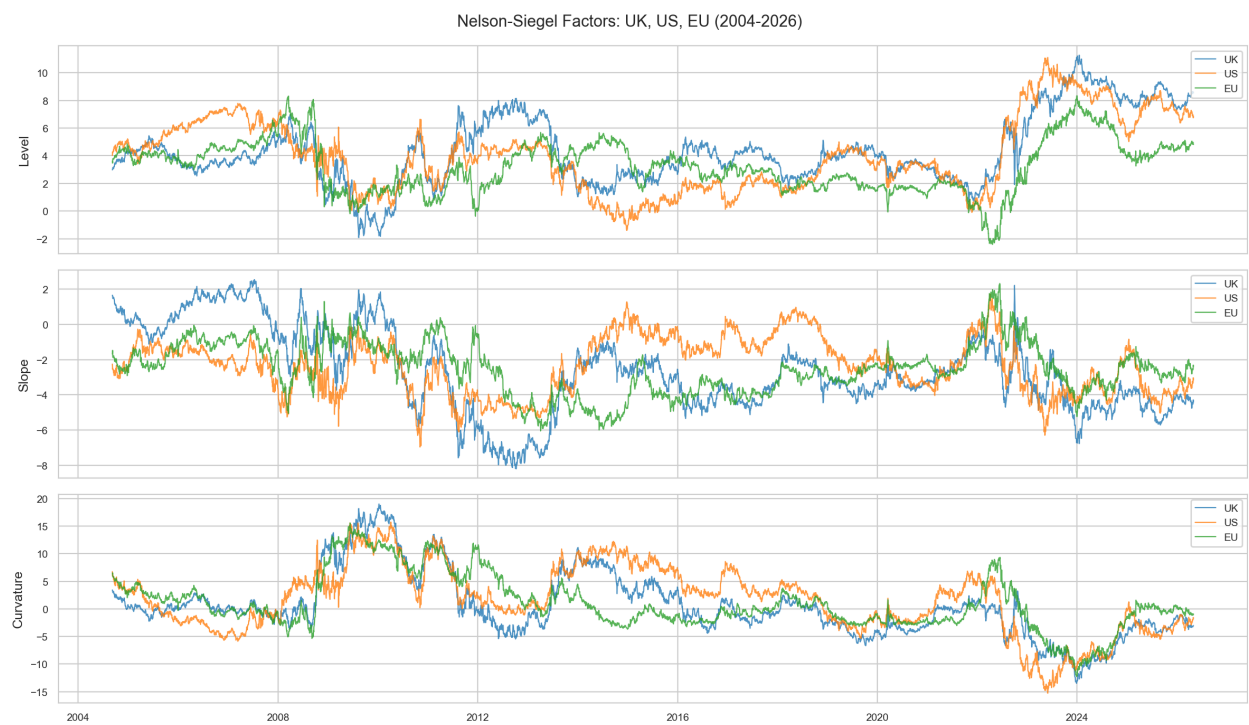


Figure 1: Nelson-Siegel factors for UK, US, EU (2004–2026). The Level factor co-moves tightly across regions; Slope diverges during different monetary policy cycles; Curvature is noisier but shows global risk-on/risk-off episodes.

Shock Propagation: How Foreign Rate Shocks Hit the UK Curve

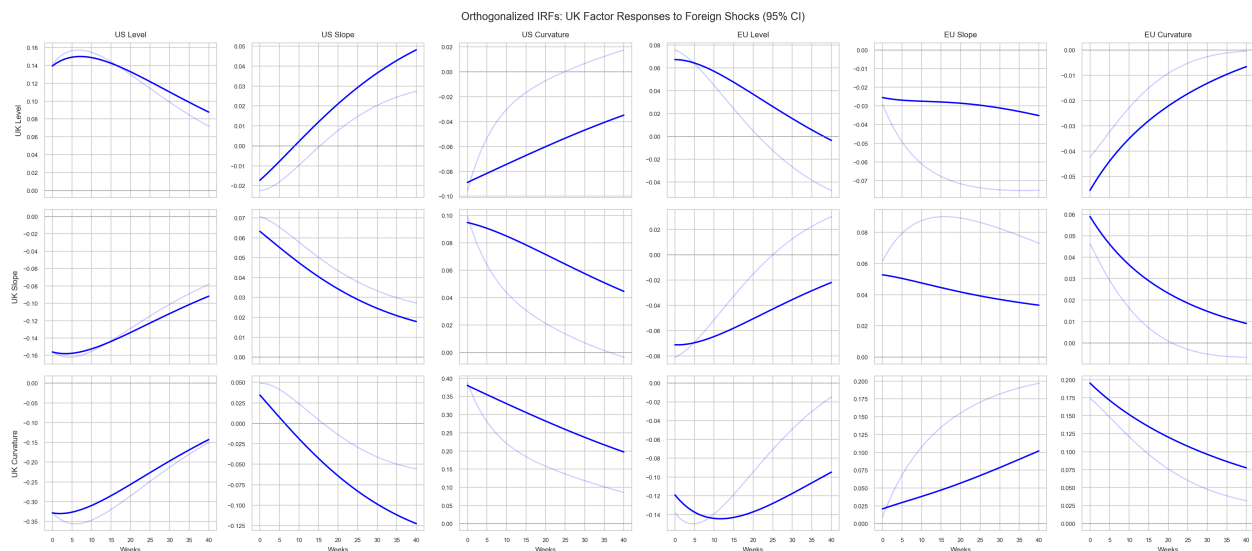


Figure 2: Orthogonalised IRFs: each panel shows the response of a UK factor (row) to a 1-s.d. shock in a foreign factor (column) over 40 weeks. Blue shading = 95% bootstrap CI.

Reading the IRF grid:

- **US Level → UK Level:** A shock to long-run US rates raises UK long rates by ~ 0.15 s.d. on impact, persisting for 30+ weeks. This is the “global real rate” channel—Treasury sell-offs transmit directly to gilts.
- **US Curvature → UK Curvature:** The strongest single spillover. A US term-premium shock propagates immediately and almost 1-for-1 to the UK belly of the curve, reflecting shared risk appetite and carry positioning.
- **EU → UK:** Weaker individually but statistically significant on impact for Level and Curvature. EU shocks decay faster than US shocks (fully absorbed within 15 weeks).
- **Slope spillovers:** Relatively muted. Monetary policy stance (slope) remains largely idiosyncratic—BoE, Fed, and ECB diverge.

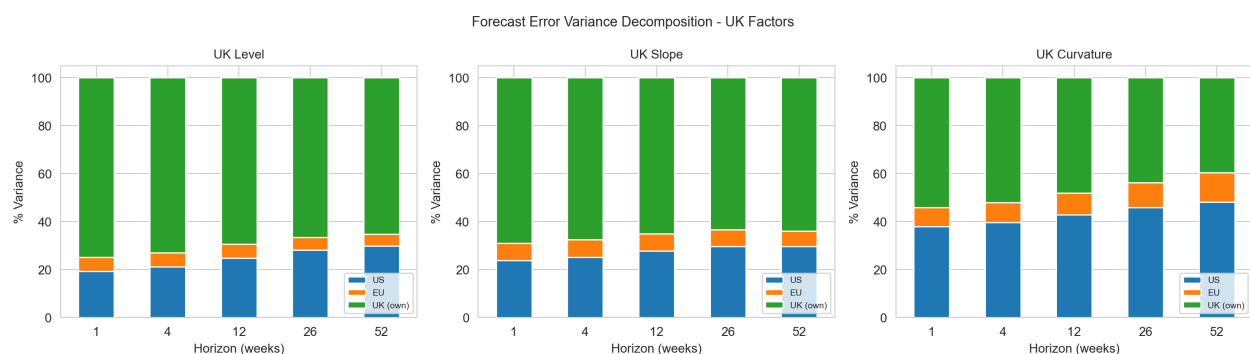


Figure 3: Forecast Error Variance Decomposition by region. At 52 weeks, foreign shocks explain 35% of UK Level variance, 36% of Slope, and 60% of Curvature.

FEVD Summary at 52-Week Horizon

	US share (%)	EU share (%)	UK own (%)
UK Level	29.8	4.9	65.2
UK Slope	29.7	6.5	63.8
UK Curvature	48.2	12.3	39.5

Key takeaway: The UK curve is *not* insulated from global moves. At the 1-year horizon, nearly half of the variation in UK term premia (curvature) is driven by US Treasuries. For trading, this means: large UST sell-offs (bear steepeners, duration supply shocks) mechanically propagate to gilts, particularly in the 5Y–10Y belly.

UK Gilt Market Context: May 2026

Macro & Fiscal Backdrop

- **Inflation:** CPI at 3.3% (March 2026), up from 3.0% in February. Energy-driven (Iran war oil shock, Ofgem cap ↑20% from July). BoE projects 3.0–3.5% through Q3 2026.
- **Growth:** GDP grew 0.5% in the three months to February 2026; OBR forecasts full-year 2026 growth at 1.1% (down from 1.4% in 2025).
- **BoE MPC (30 April):** Held at 3.75%, 8–1 vote (1 member voted for a hike). Markets now price *no cuts* in 2026, with ~3 hikes priced.
- **Gilt issuance:** DMO remit for 2026–27 set at £252.1bn (above consensus of £245bn). Heavy tilt to short/medium maturities. Revised upward on 23 April 2026.
- **Debt:** PSND at 94.3% of GDP, forecast to rise to 96.1%. OBR flags “fundamentally unsustainable” trajectory without further consolidation.

Recent Market Price Action

- 10Y gilt yield: **5.06%** (30 Apr), highest since July 2008.
- 30Y gilt yield: **5.74%** (5 May), highest since 1998 — up 13bp on the day.
- The sell-off accelerated after the MPC hawkish dissent and municipal election uncertainty.
- The April US–Iran ceasefire temporarily pulled yields 15–20bp lower; unwinding of ceasefire optimism is now reversing that move.

Upcoming Data & Events (May–June 2026)

Date	Event	Impact
8 May	UK local election results	Political risk / Labour fiscal credibility
14 May	ONS CPI (April)	Energy pass-through; key for BoE expectations
15 May	ONS Labour Market (Mar)	Wage growth vs. target
22 May	ONS GDP monthly (March)	Activity vs. rate hike risk
Late May	DMO gilt syndication (long-end)	Supply pressure on 20Y+
18 June	BoE MPC decision	Next rate decision — hike vs. hold

Key Drivers Ahead

1. **US Treasury moves remain the dominant risk factor.** Our VAR shows 30%+ of UK Level variance is explained by USTs. Any further UST sell-off (e.g., on US fiscal concerns or Fed repricing) will mechanically widen gilt yields.
2. **Energy prices / Iran situation.** A re-escalation would push UK CPI above 3.5%, forcing BoE hikes and steepening the curve. Ceasefire durability is the swing factor.
3. **Gilt supply.** The £252bn remit is front-loaded with auctions. Poor bid-to-cover ratios would signal absorption stress, particularly in the 10–20Y sector where foreign spillover is highest.
4. **BoE rate path.** The 8–1 split suggests a June hike is live. Hawkish repricing would flatten the slope (short-end up, long-end anchored by growth concerns).
5. **Fiscal credibility.** NIESR flags gilt market “derailment” risk. Any revision to spending plans or OBR downgrades could trigger a repeat of the 2022 LDI episode in the long end.

Data sources: Bank of England IADB (UK spot curve), FRED (US Treasuries), ECB Statistical Data Warehouse (EU AAA). Model: VAR(1) on weekly Nelson-Siegel factors, sample Sep 2004 – Apr 2026 (1,130 obs). This document is for informational purposes only and does not constitute investment advice.